

Certified randomness in quantum physics

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The concept of randomness plays an important part in many disciplines. On the one hand, the question of whether random processes exist is fundamental for our understanding of nature. On the other, randomness is a resource for cryptography, algorithms and simulations. Standard methods for generating randomness rely on assumptions about the devices that are often not valid in practice. However, quantum technologies enable new methods for generating certified randomness, based on the violation of Bell inequalities. These methods are referred to as device-independent because they do not rely on any modelling of the devices. Here we review efforts to design device-independent randomness generators and the associated challenges.

Because of its importance, much scientific effort is devoted to understanding when a given process generates ‘good’ randomness. A process is represented by a device, or black box, producing bits (Fig. 1), to which the user has access. What constitutes ‘good’ randomness may depend on the application, but here we are interested in the strongest definition: N bits are perfectly random if they are unpredictable, not only by the user of the device, but by any observer. This definition is satisfactory both from a fundamental and applied perspective. Although unpredictability by any observer may not be needed for some applications, such as Monte Carlo simulations, from a fundamental perspective it is difficult to argue that a process is random if there could exist an observer able to predict its outcomes. The requirement that the results should be unpredictable by any observer guarantees that the generated randomness is private: the user, by running the process in a secure location, can be certain that nobody knows the obtained results. Such private randomness can safely be used for cryptographic purposes.

According to our definition, the generation of randomness from scratch is impossible. This follows from the unfalsifiable hypothesis of the existence of a super-deterministic model in which everything, including the entire history of our Universe, was pre-determined in advance and known by the external observer. Therefore, any protocol for randomness generation must be based on some hypotheses or assumptions. The appropriateness of the assumptions is often debatable and strongly depends on the application. From a fundamental point of view, one random number generator (RNG) is better than another if it is based on fewer or weaker assumptions. However, adding more assumptions may not compromise the use of the random bits for a specific application, while it may simplify the experimental implementation and/or increase the efficiency.

Here we adopt a physics-based approach to randomness generation: the random numbers should be unpredictable by any physical observer, that is, any observer whose actions are constrained by the laws of physics. In particular, and according to the current understanding of nature, the device generating the random numbers and the device held by the potential adversary should obey the laws of quantum physics. Within this theory, the joint state of the user U and the adversary E (which includes the environment) describing N ideal random bits is

$$\rho_{UE} = \left(\frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| \right)^{\otimes N} \otimes \rho_E \quad (1)$$

This state corresponds to N realizations of a perfect random bit, which takes the value 0 and 1 with equal probability and is totally uncorrelated with the state of the adversary, or environment, ρ_E . We say that

a given device generates ‘arbitrarily good randomness’ if its output is undistinguishable from the ideal state given in equation (1), up to some controllable small error.

A fundamental issue when considering randomness generation is certification: how can the user certify that the numbers produced by their device are random? According to equation (1), the random bits should follow a uniform probability distribution and be uncorrelated to the environment. Concerning the first point, the standard method for certifying randomness consists of running statistical tests¹ on sequences generated by the device. However, it is unclear what passing these tests means and, in fact, it is impossible to certify with finite computational power that a given sequence is random.

The certification of privacy is much more subtle. The best way to illustrate this is by means of what we call the memory-stick attack. Imagine a situation in which the provider of the devices is the adversary and has access to a proper RNG. The provider uses this RNG to generate a long sequence of good random numbers, stores them into a memory stick and sells it as a proper RNG to the user. The numbers generated by the user will pass any statistical test and look random; however, they are not properly random because they can be perfectly predicted by the adversary.

The generation of good randomness is a notoriously difficult problem^{2,3}. Basically, there are three types of approach. First, pseudo-random-number generators (PRNGs) use an algorithm to process an initial random seed. They are fast, cheap and the properties of the generated sequences are good enough for some applications. However, the random character of the output of PRNGs and their privacy are based on assumptions about the computational power of the adversary. But this is not the criterion adopted here—we demand unpredictability by any observer, independently of its computational power.

The second type of RNG is called a true RNG (TRNG) and exploits physical processes that are hard to predict, such as meteorological phenomena or the mouse movements of a computer user. Finally, there are quantum RNGs (QRNGs), which exploit a quantum process believed to be fundamentally random. In what follows, we focus the discussion on QRNGs, although many of the problems also apply to TRNGs.

The paradigmatic example of a QRNG is defined by the clicks observed after single photons impinge a beam-splitter; see Fig. 1. However, this idealized set-up may be difficult to implement in an experiment. More generally, imperfections on the devices are unavoidable and could deteriorate the quality of the generated numbers in uncontrolled manners. It is also difficult to exclude memory effects, for instance at the detectors, which produce correlations among the generated bits. The privacy of the symbols follows from the fact that the single-photon state is assumed to

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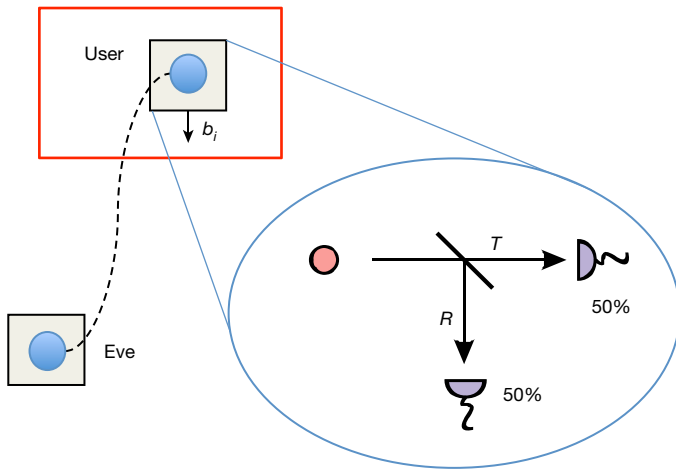


Figure 1 | Schemes for randomness generation. The user, in their secure location, has access to a device that generates bits b_i . The user wants to make sure that the value of these bits cannot be predicted by any observer outside their laboratory. The way to model this is by an external ‘super-observer’ who has access to any physical system other than that at the user’s location, which we represent by another device that may be correlated to the user’s. It is useful to interpret the external observer as an adversary or eavesdropper, ‘Eve’, who wants to predict the generated bits (for instance to hack any possible use for cryptographic applications). The generated bits should be unpredictable by Eve, even after measuring their device. For standard QRNGs, the random character of the outputs follows from assumptions about the inner working of the user’s device. The figure displays a scheme based on a single photon (red ball) impinging a beam-splitter with a transmission coefficient T , and reflection coefficient R , of $1/2$. Two single-photon detectors placed at the two arms of the interferometer measure the path taken by the photon. According to quantum physics, this process is probabilistic and the probability that a given detector ‘clicks’ is equal to the transmission coefficient of the beam-splitter, which is assumed to be $1/2$.

be pure and therefore cannot be coupled to another system. Hence, there are many assumptions about the working of the QRNG that are crucial to guarantee the perfect match between the ideal theoretical situation and the implementation, which is in turn necessary to guarantee the quality of the generated outputs.

The current certification method applied to QRNGs consists of passing statistical tests. Apart from the problems already mentioned, the use of these tests is even more problematic for QRNGs because they can also be satisfied by classical RNGs. So, the only guarantee the user has that the symbols have a quantum origin is the word of the provider. Furthermore, the user has no means to test the privacy of the symbols because they cannot rule out the memory-stick attack. Trusting the provider is again the only option. This level of trust between provider and user is unsatisfactory because in many situations, especially for cryptographic applications, it is convenient to reduce the trust on the provider as much as possible. And, even if the provider is trusted and has constructed the devices in the best possible way, uncontrolled drifts and changes on the devices are unavoidable and could deteriorate the quality of the generated randomness. Consequently, there is a need for solutions that certify the ‘quantumness’, quality and privacy of QRNGs without requiring any detailed modelling of the devices.

Device-independent QRNGs (DIQRNGs) offer a solution to the previous issues and provide protocols for generating certified randomness based on only general assumptions about the set-up, such as the validity of quantum physics. In particular, they do not require any assumptions about the inner working of the devices, which can be seen as quantum black boxes processing classical information. The development of DIQRNG protocols is an active research field involving many concepts and methods, including information-theoretical studies to design stronger protocols based on weaker assumptions and their experimental realization using current or near-future technology.

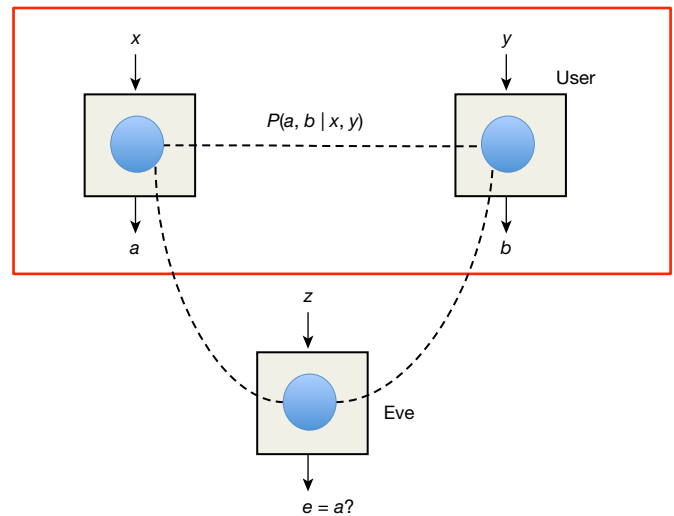


Figure 2 | Structure of DIQRNG protocols. In a general protocol for a DIQRNG, the user has access to $n \geq 2$ correlated devices. The figure shows the simplest case of two devices (boxes inside the red rectangle), which generate classical outputs a and b after applying the inputs x and y . The inputs x and y can be understood as the labels of the measurements performed on each device and the outputs as the obtained results. The external (eavesdropping) observer, Eve, may have a system that is correlated with the user’s devices. The randomness of one of the outputs, say a , can be quantified by the optimal probability P_{guess} that Eve guesses it correctly ($e = a$) after performing a measurement z on their system^{14,15,77}. In the case of quantum eavesdroppers, the guessing probability is optimized over all quantum preparations, including the tripartite state and measurements, which are compatible with the correlations observed by the user^{15,88–90}. The assumption of the validity of quantum mechanics can be relaxed by considering eavesdroppers who can prepare any tripartite correlations that are compatible with the no-signalling principle, even those that are incompatible with quantum physics^{15,90}.

In what follows, we first show how randomness certification without assumptions about the inner working of the devices can be achieved by exploiting the quantum violation of Bell inequalities. We then describe the state of the art in DIQRNG protocols and their experimental implementations. Because these implementations turn out to be challenging, we describe other approaches to certified randomness with milder experimental requirements. We also explain how, in addition to practical applications, protocols for certified randomness answer some fundamental questions in physics. We conclude with an outlook.

Device-independent randomness generation

DIQRNG protocols make use of the correlations observed when measuring entangled particles that do not have a classical analogue, as certified by the violation of a Bell inequality⁴. For such protocols, the user needs at least two separated devices to run a Bell test; see Fig. 2. The devices receive classical inputs x and y , and produce classical outputs a and b . After N_r rounds of collecting the data (x, y, a, b) , the user calculates the relative frequencies of the outcomes given the inputs $P(a, b | x, y)$, which can be estimated without making any assumptions about the internal working of the devices. A Bell inequality is a linear function of these relative frequencies

$$\beta = \sum c_{abxy} P(a, b | x, y) \leq \beta_L$$

characterized by the coefficients c_{abxy} . Here, β_L is the local bound satisfied by classical theories, such as that of Einstein–Podolsky–Rosen (EPR)⁵. The violation of the Bell inequality indicates the presence of non-classical correlations between the two devices.

The idea behind DIQRNGs is that, if the user observes a Bell inequality violation, they can be certain that the unknown quantum state in the devices is entangled and pure. The purity of the quantum state certifies

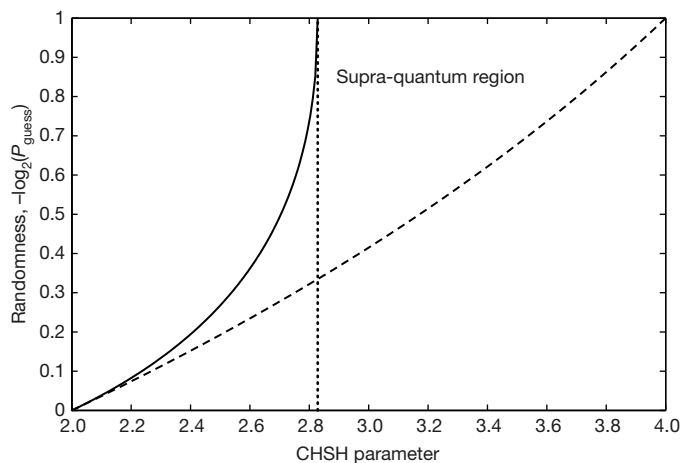


Figure 3 | Randomness as a function of the CHSH Bell inequality violation observed by the user. The amount of generated randomness at one of the outputs is shown, expressed as $-\log_2(P_{\text{guess}})$, where P_{guess} is the optimal guessing probability of the eavesdropper, such that a value of 0 corresponds to an adversary being able to correctly predict all outputs and a value of 1 to a perfectly random bit (for which the adversary will correctly guess the outputs half the time). The CHSH Bell inequality violation is expressed in terms of the CHSH parameter, with a value of 2 corresponding to no violation (the local bound). The curves are computed using the techniques in refs 15 and 77. The solid line refers to a quantum eavesdropper; the dashed line is for a non-signalling eavesdropper. The quantum violation of the CHSH inequality is upper-bounded by $2\sqrt{2}$ (vertical dotted line), but it is possible to get larger, ‘supra-quantum’ violations without breaking the no-signalling principle. At the local bound (at which the CHSH parameter is equal to 2), no randomness can be certified, whereas some randomness appears for any violation, that is, when the CHSH parameter is larger than 2. In both cases, a perfect random bit is certified by the corresponding maximal Bell violation, which corresponds to a CHSH parameter of $2\sqrt{2}$ in the quantum case and 4 for non-signalling correlations.

that the two devices are not too correlated with the environment or the external observer. The entanglement certifies that the local state of one of the devices is mixed and, thus, a measurement on it generates random outcomes. Moreover, the Bell certification of randomness is intrinsically quantum, because classical devices always satisfy a Bell inequality. Finally, the randomness is device-independent because only the observed statistics $P(a, b|x, y)$ are needed for its computation.

Consider a case in which the user tests the Clauser–Horne–Shimony–Holt (CHSH) Bell inequality⁶ and observes its maximal quantum violation¹. (Although some of the following results are illustrated using the CHSH inequality—the simplest Bell inequality—the main ideas and concepts apply to any Bell inequality.) According to quantum physics, the statistics of local measurements $\Pi_{a|x}$ and $\Pi_{b|y}$ on the state describing the two quantum devices $|\Psi\rangle$ are given by

$$P(a, b|x, y) = \langle \Psi | \Pi_{a|x} \otimes \Pi_{b|y} | \Psi \rangle \quad (2)$$

A maximal violation of the CHSH inequality is obtained only when projective measurements are performed on a maximally entangled state of two qubits^{7–9}

$$|\psi\rangle = |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Because the state is pure, it cannot be correlated to the environment. The local measurements on half of it produce perfect random bits according to equation (1), which are certified by the observed Bell violation. This intuitive argument holds when the maximal quantum violation of the CHSH inequality is obtained. For noisy, non-maximal violations, it is also possible to quantify the amount of randomness from the observed violation; see Fig. 3.

Although the previous discussion has exploited the properties of quantum correlations described by equation (2), it is possible to design DIQRNGs that do not rely on the validity of quantum theory, but on only that of the no-signalling principle, that is, the impossibility of faster-than-light communication between devices. Under the sole assumption of no-signalling, the violation of a Bell inequality guarantees the random character of the outputs^{10–16}.

All these nice features come at a price: the user should meet the conditions needed in equation (2). They must ensure that:

- (C1) the inputs (x, y) have no correlations with the devices; and
- (C2) there is no communication between the two devices during the generation of the two distant outcomes (see Fig. 2).

Looking at equation (2), condition (C1) implies, for example, that the measured quantum state is independent of x and y . Condition (C2) imposes the tensor product and that the measurements on one device do not depend on the input of the other. To guarantee these conditions, the user needs to make physical assumptions about the devices (albeit not about their internal working).

Protocols

We now provide a unified description of most of the protocols for DIQRNGs that have been proposed so far. In general, these protocols involve $n \geq 2$ devices, each having an input x_i and an output a_i for $i = 1, \dots, n$. Condition (C1) imposes that the inputs $(x_1, \dots, x_n)$ must be selected in a way that is uncorrelated to the devices. A standard (but not the only) way of satisfying this condition is by choosing the inputs using a random seed, which has to meet certain ‘independence’ requirements depending on the protocol (see Box 1).

A series of works have focused on efficiency, trying to optimize the trade-off between initial and final randomness. The corresponding protocols are known as randomness expansion protocols^{14,15,17–23}. It has been proven that an unbounded amount of randomness (length of the final random bit string $N_k \rightarrow \infty$) can be generated from a finite seed^{19,21}.

Other works^{19,21,24–29} have focused on the quality of the seed and studied how arbitrarily good randomness can be generated in Bell set-ups using sources of imperfect randomness. These protocols are often known as ‘randomness amplification’ protocols²⁴. A commonly used model for the imperfect seed $(s_1, s_2, \dots)$ is a Santha–Vazirani source³⁰, which is characterized by a parameter $\epsilon \in (0, 1/2)$ such that

$$\epsilon \leq P(s_i | s_1, s_2, \dots, s_{i-1}, \text{devices, Eve}) \leq 1 - \epsilon \quad (3)$$

for all i , in which ‘Eve’ denotes any correlated information owned by the eavesdropper. Therefore, the larger the value of ϵ , the higher the randomness of the bits. Other models for the seed, more general than the Santha–Vazirani source, have also been considered, such as min-entropy sources^{21,27}. In the case of Santha–Vazirani (equation (3)), the performance of randomness amplification protocols is measured by comparing the parameter ϵ_i of the initial source with the final ϵ_f of the generated bits. Full randomness amplification is attained when a source with $\epsilon_i \rightarrow 0$ is mapped to one with $\epsilon_f \rightarrow 1/2$ (ref. 29). From a fundamental point of view, the existence of protocols that attain full randomness amplification is important; we return to this point below. However, from a practical point of view, although allowing for a non-perfect seed is a good addition to the security of a protocol, in most applications one can assume that the seed is uncorrelated to the adversary and devices, in which case the expansion rate is a more practical figure of merit. Randomness expansion and amplification protocols are asides of the general problem, which is the generation of device-independent private randomness under the minimal set of assumptions and with minimal resources (see Table 1).

Bell-certified randomness is also a resource for device-independent quantum key distribution^{11,13,23,31–42}. However, the goal here is not only to generate randomness, but to establish a secret key between two distant users using a Bell violation observed between their devices. In particular, and in contrast to the case of randomness generation, the devices are held

BOX 1

DIQRNG PROTOCOL

Steps of the protocol

Data collection. Repeat steps (a), (b), (c) N_r times:

(a) A source of n -partite entangled states sends a particle to each of the n devices.

(b) Part of the seed is processed to generate a sample from the prior distribution $P(x_1, \dots, x_n)$ of the inputs applied to each device. This distribution can be optimized before implementing the protocol to suit the statistics of the devices.

(c) Measurement x_i is performed on device i generating outcome a_i . The inputs and outputs of the devices ($a_1, \dots, a_n, x_1, \dots, x_n$) are stored.

Non-locality estimation. Calculate the relative frequency of every combination of inputs and outputs $P_{\text{freq}}(a_1, \dots, a_n | x_1, \dots, x_n)$ using the raw data collected in the N_r rounds. From this data, estimate the non-locality of the observed correlations and determine the length N_k of the final random bit string. The larger the amount of non-locality, the longer N_k . If the observed non-locality is insufficient, then the protocol is aborted ($N_k = 0$).

Classical post-processing. Generate the final N_k -bit string using the raw data collected in the N_r rounds plus an additional part of the seed. This process is often made with a randomness ‘extractor’^{91,92}.

Parameters of the protocol

The parameters that are relevant to the design of DIQRNG protocols are as follows.

Efficiency. This parameter describes the trade-off between the amount of randomness generated and the resources consumed by the DIQRNG protocol. Examples of these resources are the number of random bits of the seed N_s or the number of times the devices are used N_r .

Quality of the seed. The random seed may not be perfect. For example, the seed may not necessarily be uniformly distributed, or display correlations with the devices or the adversary. Instead of using a seed, another possibility is to assume the existence of a good and free public source of randomness, such as the broadcast by the National Institute of Standards and Technology (NIST) Randomness Beacon (http://www.nist.gov/itl/csd/ct/nist_beacon.cfm). In this case, the protocol generates private randomness from public randomness.

Robustness. This parameter characterizes the tolerance of the protocol to noise and imperfections. This allows for using realistic noisy apparatuses. A protocol is robust if it works ($N_k > 0$) for violations above some threshold, which does not need to coincide with the local bound.

Number of devices. The minimum number of devices used in the protocol is $n = 2$.

Composability. Protocols should be such that, if the adversary learns some information about the N_k final random bits, then they should be able to deduce essentially no additional information⁹³.

Physical assumptions. Many protocols assume that all the devices and the adversary are constrained by the laws of quantum mechanics. However, it is possible to relax this requirement and assume only the validity of the no-signalling principle. Even when the security of the protocol does not rely on the validity of quantum mechanics, quantum technology is still needed to generate Bell-violating correlations.

in two separate locations and the channel between them is accessible to the eavesdropper.

Implementations

The implementation of the aforementioned DIQRNG protocols requires the observation of a Bell inequality violation. For that, it is necessary to prepare an entangled state of $n \geq 2$ particles, which are distributed to n devices where they are subjected to local measurements. Assuming the validity of quantum physics, the experimental set-up should guarantee that conditions (C1) and (C2) are met so that the observed statistics are correctly described by equation (2).

An experimental challenge for the observation of Bell inequality violations is that a high detection efficiency (more than about 70%) is required to close the detection loophole⁴³. This loophole implies that, for low enough detection efficiencies, the statistics of a Bell experiment can always be described by an EPR model, which is deterministic, and so no randomness certification is possible^{44,45}. Closing the detection loophole is demanding because it concerns any losses in the set-up.

Table 1 | State-of-the-art DIQRNG protocols

	n	$N_k(N_s)$	N_k/N_r	Seed quality	Seed privacy	QM
Ref. 21	Large	Infinite	0	$\epsilon > 0$	No	Yes
Ref. 20	2	Exponential	> 0	$\epsilon = 1/2$	Yes	Yes
Ref. 28	2	Small	0	$\epsilon > 0$	Yes	No
Ideal	2	Infinite	> 0	$\epsilon > 0$	No	No

Properties of the best known protocols, which are robust, composable and immune to attacks that exploit memory effects on the devices, as in the memory loophole introduced in the context of Bell inequality violations⁸⁷. The parameters are: the number of devices n ; the amount of expansion of the initial seed $N_k(N_s)$, expressed as the way in which the length of the final random bit string N_k depends on that of the initial seed N_s ; the efficiency rate N_k/N_r , where N_r is the number of times the devices are used; the seed quality ϵ in terms of equation (3); whether publishing the seed compromises security; and whether the security of the protocol relies on the validity of quantum mechanics (QM). Note that infinite $N_k(N_s)$ assumes that N_s is sufficiently long. The bottom row contains the optimal value for each parameter.

But it is a loophole that can be, and actually has been^{46–53}, completely closed, because it does not put into question the validity of the description given in equation (2), but simply demands a high enough detection efficiency.

Contrary to the detection loophole, the locality⁵⁴, collapse-locality⁵⁵ and free-will⁵⁶ loopholes do put into question the validity of equation (2). Consequently, they can never be strictly closed, but their plausibility can be enforced by making physically motivated assumptions about the experimental arrangement. The locality loophole affects condition (C2), that is, whether the measurements in either device depend on what happens on the other device. The standard approach to closing this loophole is based on Einstein’s theory of relativity and involves arranging the measurements so that they define events that are separated in a space-like way such that no communication can take place between the two devices. This is a very satisfactory solution, but also a demanding one. In our view, there are relevant scenarios in which it is also possible to assume the validity of (C2) without space-like-separated measurements. For example, a low level of trust may be put on the provider so that it is safe to assume that the devices do not signal to each other when producing the outputs given the inputs. Or some level of shielding, essential for any cryptographic use of the generated numbers, may be assumed, which can in turn be used to avoid any unwanted communication between the devices.

Moreover, the space-like arrangement that is usually adopted to close the locality loophole also assumes that there is precise knowledge about when the local measurements start and end, that is, when the inputs x, y are defined and the outputs a, b are produced. This issue relates to the free-will and collapse-locality loopholes, which also put into question condition (C1). If there is no timing information about when the inputs are generated, then this information could, for instance, exist before the entangled state is produced; that is, the state in equation (2) could depend on x and y . A proposed theoretical way to avoid this dependence is to assume that the inputs are generated by humans, hence the term ‘free will’. The usual and more practical approach to close the loophole is to use a standard QRNG⁵⁶. However, it is questionable whether (and if so, why) a QRNG is preferable over other processes of classical origin⁵⁷. Similar considerations apply to the timing of the outputs, as in the collapse-locality loophole: one needs to define when the classical results are actually produced to guarantee that the measurements define space-like-separated events.

Taking into account all of these points and the technological state of the art, two set-ups emerge in which to implement DIQRNG protocols: entangled particles in separate locations and entangled photons. In the first case, a Bell test is performed between two distant massive particles, such as nitrogen-vacancy centres⁵¹, or ions in two traps^{47,48} that have been entangled through entanglement swapping on two photon–particle entangled pairs^{58,59}. The advantage of this set-up is that massive particles can be measured with almost perfect efficiency, thus closing the detection loophole. A first proof-of-principle demonstration of a DIQRNG, reporting a generation rate of 42 random bits after approximately one month of measurements, was performed using two entangled ions in two traps at 1-m distance¹⁵. This rate is valid under the assumption that the experimental set-up was not operating in a malicious way¹⁷. Although

challenging, set-ups with entangled distant particles also allow the measurements to be arranged so that one can reasonably assume that both the detection and locality loopholes are closed. This was achieved in ref. 51 and subsequently in ref. 60; however, these experiments do not report any analysis of random-number generation.

In the second case, the set-up consists of performing polarization measurements on two entangled photons. Historically, one of the main challenges in these set-ups was that photon-detection efficiencies were too low to close the detection loophole; however, advances in photo-detectors have allowed it to be closed^{49,50}. The first experiment focused on only the Bell violation, but the second reported a random-number generation rate of 0.4 bits s^{-1} . More recently, the locality loophole has also been closed in photonic experiments^{52,53}; but again, none of these experiments was analysed in the context of device-independent random-number generation.

Other methods for randomness generation

Because meeting conditions (C1) and (C2) in an experiment is challenging, alternative proposals for certified quantum randomness generation have been proposed. The idea is to keep part of the device-independent spirit and make only mild assumptions about the set-up, yet without any detailed modelling of the devices. Randomness certification comes from a purely quantum effect with no classical analogue, under the mentioned assumptions. Standard QRNGs do not fit into this category because they require modelling and certify randomness using statistical tests that are also satisfied by classical RNGs.

A series of works have explored information protocols under a dimensional constraint^{61,62}—a scenario known as semi-device-independent. The set-up is different from a Bell test and consists of a preparing device that prepares a system in different quantum states and a measuring device that performs measurements on it. It is assumed that the states prepared by the first device and measured by the second belong to a Hilbert space of dimension not larger than d ; this is the extra assumption that goes beyond the fully device-independent paradigm. Randomness certification is then obtained via the violation of the so-called dimension witnesses⁶³. One of the practical advantages of this approach is that it does not require the generation of entanglement. The required detection efficiencies are smaller, but still demanding to achieve experimentally⁴⁵. A solution to this problem was suggested in ref. 64, where it was shown that schemes in which one assumes that the preparation and measuring device share no correlations, and that devices do not display memory effects, certify the presence of randomness for any value of the detection efficiency. A proof-of-principle experimental demonstration of this proposal has been performed⁶⁵, but a security proof for these schemes without assumptions about memory effects is lacking.

A second proposal considers an asymmetric scenario in which some of the devices are fully trusted; for example, one may trust the preparing devices, but not the measuring devices. Asymmetric scenarios are often considered in the context of steering⁶⁶. This concept is defined in the same set-up as non-locality, in which two parties perform measurements on two distant quantum particles, but now one of the devices is fully trusted. The detection of steering provides a quantum certification that is sufficient to guarantee the presence of randomness⁶⁷. Steering has been experimentally demonstrated with the detection and locality loopholes closed^{68–70}, but the detection efficiencies needed for randomness expansion still pose a challenge⁴⁵. Security proofs are in a preliminary stage also in the case of steering. A general security proof was provided in ref. 71, but it requires a very low level of noise.

Fundamental questions on randomness

The results connecting randomness and non-locality are relevant not only for applications of quantum technologies, but also for our understanding of physics. Protocols that attain full randomness amplification against non-signalling eavesdroppers represent the strongest form of certification using quantum physics of the existence of random events in nature. It is impossible to certify randomness from scratch. Under only

the assumption of no-signalling, the violation of Bell inequalities certifies the presence of randomness, but also requires some initial randomness. Full randomness amplification protocols^{19,21,25–29} are not able to completely break this circularity, but relax it as much as possible.

Historically, the discussion on EPR models and Bell inequalities was motivated by the search for a ‘complete’ alternative to quantum theory, in the sense that measurement outcomes could have a deterministic description within the alternative theory. The violation of Bell inequalities implies that quantum predictions cannot be reproduced by a deterministic theory without violating the no-signalling principle. In recent years, stronger proofs of the ‘uncompleteness’ of quantum theory have appeared^{11,13,72–74}. These works show that a no-signalling model, possibly non-deterministic, that has higher predictive power than quantum theory does not exist. Full randomness amplification protocols against no-signalling eavesdroppers can also be seen as proofs of the uncompleteness of quantum theory. The study of randomness is also relevant when comparing quantum theory with more general theories that respect the no-signalling principle^{9,75}. It has been shown that theories leading to general non-signalling correlations do not allow for maximal randomness certification, whereas quantum theory does⁷⁶.

Finally, a series of works have shown that the relation between entanglement, non-locality and randomness is more subtle than expected. For instance, states with arbitrarily small amounts of entanglement (and non-locality) allow for maximal randomness certification⁷⁷. Recent progress in this direction shows that the use of more complex measurements, such as non-projective⁷⁸ or sequences of measurements⁷⁹, provides further advantages for randomness certification. Despite these results, a complete understanding of the relation between entanglement, non-locality and randomness is still missing.

Outlook

DIQRNG protocols address fundamental and practical drawbacks of standard RNG schemes and so represent a change of paradigm for randomness generation. On the theory side, the existing security proofs show the validity of the approach. However, further theoretical studies are needed to understand how to relax the requirements for DIQRNGs. The ultimate goal would be to design a robust and composable secure protocol that attains an infinite randomness expansion rate using initial sources of arbitrarily weak public randomness with only two devices, and assuming only the validity of the no-signalling principle (see Table 1). Although this ambitious goal may be unreachable, there is still a lot of room for improvement on the conditions that are needed for device-independent randomness generation.

On the implementation side, it is expected that new DIQRNG experiments using the set-ups explained above will be reported in the coming years, with constantly improving generation rates. Looking ahead, integrated photonic circuits and solid-state set-ups present as other platforms in which to run the previous protocols. To our knowledge, however, no Bell experiment has been reported on integrated photonic circuits. Non-local correlations in solid-state devices, as certified by the violation of a Bell inequality, have been reported for two-^{80,81} and three-component systems^{82,83}. These technologies could be more promising in terms of miniaturization and, hence, the possible construction of commercial DIQRNGs. However, miniaturization comes at a price: the possible validity of condition (C2) becomes less clear, even if one accepts that measurements do not need to be space-like separated. Theoretical solutions to take into account cross-talk effects have been proposed⁸⁴. In fact, the analysis of implementations of DIQRNGs leads to new theoretical questions, such as being able to determine: (i) the detection efficiencies required for randomness expansion⁴⁵; (ii) which Bell set-ups are more robust against noise or detection inefficiencies⁸⁵; and (iii) the ways in which to deal with detection inefficiencies⁸⁶.

Continued advances in the theory and implementation of DIQRNGs will lead to the design of more robust and feasible schemes. RNGs with unprecedented standards of quality and security seem within reach using quantum technologies.

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